CTC and ACTC Participation Results and IRS-Census Match Methodology, Tax Year 2021

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Abstract

The Child Tax Credit (CTC) and Additional Child Tax Credit (ACTC) help ease the financial burden of families with children. This paper provides taxpayer and dollar participation estimates for the CTC and ACTC covering tax year 2021. In this tax year the credit was temporarily expanded under the American Rescue Plan Act. The estimates derive from linking the 2022 Current Population Survey Annual Social and Economic Supplement (CPS ASEC) to IRS administrative data. This approach, called the Exact Match, uses survey data to identify CTC/ACTC eligible taxpayers and IRS administrative data to indicate which eligible taxpayers claimed and received the credit. Overall, in tax year 2021, eligible taxpayers participated in the CTC and ACTC program at a rate of 91 percent while dollar participation was 85 percent.

Keyword: Child Tax Credit, Additional Child Tax Credit, Participation

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[*] Any opinions and conclusions expressed herein are those of the authors and do not represent the views of the U.S. Census Bureau. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 6000463: CBDRB-FY25-CES004-030, CBDRB-FY26-CES004-009, CBDRB-FY26-CES004-010). The views expressed in this paper are those of the authors and do not necessarily represent the views of the IRS.

Introduction and Background

The Child Tax Credit (CTC) offers assistance to help ease the financial burden of families with children. Enacted by the Taxpayer Relief Act of 1997 (P.L. 105-34), the CTC originally offered families with children a nonrefundable credit of $400 per child beginning in 1998. The CTC is not limited to low-income taxpayers and is now sometimes refundable. The refundable portion of CTC, called the Additional Child Tax Credit (ACTC), was established in the Economic Growth and Tax Relief Reconciliation Act of 2001 (P.L. 107-16). Since 2001, the credits have seen several changes and currently operate under the One Big Beautiful Act of 2025 (P.L. 119-21). The American Rescue Plan Act of 2021 (P.L. 112-2) temporarily expanded the credit by increasing the credit amount to $3,600 for qualifying dependents age less than 6 and $3,000 for qualifying dependents ages 6-17, made the credit fully refundable, and directed half of the anticipated credit to be paid in advance monthly payments July-December 2021. The expanded credit also had different phase out ranges when compared to prior tax years. This paper provides estimates of national participation for the combined Child Tax Credit (CTC), Additional Child Tax Credit (ACTC), and Advance Child Tax Credit payments covering tax year 2021.

Data

The Exact Match methodology for estimating the participation rates consists of linking individuals in the Current Population Survey Annual Social and Economic Supplement (CPS ASEC) to their tax return information from IRS administrative data. The CPS ASEC identifies the CTC/ACTC eligible taxpayers and eligible dollars, while the IRS administrative data and return information indicate which taxpayers claimed and received the credits and dollars paid. The following sections describe the CPS ASEC and IRS administrative data and then detail the process of producing the participation rates.

Current Population Survey Annual Social and Economic Supplement

The CPS ASEC is a stratified random sample of 60,000-90,000 household addresses from the civilian, non-institutionalized population in the United States and is collected by the Census Bureau. It is a supplement to the monthly Current Population Survey, conducted in February-April of each year, containing detailed information on demographic characteristics, annual earnings and incomes, employment, hours worked, and family structure. It also serves as the official source of U.S. income and poverty statistics. All income information in the CPS ASEC refers to the prior calendar year while demographic information is as of the survey interview date. Since the CPS ASEC does not collect tax information, the National Bureau of Economic Research's (NBER) TAXSIM program is used to model tax returns, including adjusted gross income, earned

income, number of qualifying dependents, the Child Tax Credit, the Additional Child Tax Credit, and other important tax return fields.

IRS Administrative Data

The IRS provides the Census Bureau administrative tax data and return information. The return information covers all returns through cycle 26 of the year after returns are filed (e.g., tax year 2021 returns posted through cycle 26 of 2023), allowing for most audits to be completed.[1] The key variables include number of qualifying dependents for the CTC, ACTC, and Credit for Other Dependents (ODC), as well as claimed and allowed CTC and ODC amounts. Other important variables include wages, tips, and other compensation before any payroll deductions from all jobs in each tax year, adjusted gross income, investment income, and self-employment income. The Advance CTC Payments file provides all advance payments made July-December 2021, aggregated by taxpayer, not tax return.

Linking CPS ASEC to Administrative Tax Data

The Economic Reimbursable Surveys Division (ERD) at the Census Bureau processes the CPS ASEC and administrative tax data from IRS to add a unique anonymized identifier called a Protected Identification Key (PIK) that allows the two data sources to be linked. ERD assigns the PIK using proprietary linkage software, the Person Identification Validation System (PVS). PVS assigns the PIK in the administrative tax data using Social Security Number. Since the Social Security Number is not collected in the CPS ASEC, PVS assigns the PIK using a probabilistic matching model based on name, address, date of birth, and sex.[2] To make the resulting linked CPS ASEC sample representative of the U.S. population, the weights of the linked individuals are adjusted, or re-weighted, to allow for PIK non-assignment (re-weighting described later in Weighting Procedure).

Methodology

The CTC/ACTC methodology provides two primary statistics to gauge CTC/ACTC participation: (1) the taxpayer participation rate and (2) the dollar participation rate. The taxpayer participation rate (a percentage) is the total number of eligible taxpayers who received a CTC/ACTC payment divided by the total number of eligible taxpayers. The dollar participation rate (a percentage) is the total amount of eligible CTC/ACTC dollars paid out divided by the total CTC/ACTC dollars for which all taxpayers are eligible. Both participation rates are for the combined CTC and ACTC and represent participation conditional on eligibility.[3]

[1] Cycle 26 refers to the first 26 weeks of the year.

[2] For more information about the PVS system refer to Wagner and Layne (2014) and Raze, Perales, and Landivar (2026).

[3] Taxpayers who receive the credit but are not modeled eligible are not included in the participation rates.

Modeling CTC and ACTC Eligibility and Amount in the CPS ASEC

The first step of the participation methodology involves modeling tax units, CTC/ACTC eligibility, and CTC/ACTC credit amount using the CPS ASEC. The CPS ASEC does not collect any information on tax filing behavior, so we use the NBER TAXSIM microsimulation program to model the tax return, which is then used to determine CTC/ACTC eligibility.[4] An important part of this modeling assigns individuals and dependents in the CPS ASEC to tax units using household, family, and relationship information and designating the primary filer within each tax unit. The NBER TAXSIM program takes up to 35 input variables, including filing status, number of tax unit members, number and ages of dependents, earnings, and other income items, and applies federal tax rules (including rules for CTC and ACTC) to simulate a federal tax return.[5] The simulated tax return contains adjusted gross income, filing status, tax liability which is used to determine CTC and ACTC eligibility and amount.[6]

Adjustments to Original Modeled Eligibility

While the NBER TAXSIM model provides initial estimates of credit eligibility and amount based on the CPS ASEC, we further refine these initial estimates using IRS administrative data to account for tax behavior not predicted by the model and to correct for underreporting of income in the CPS ASEC. We implement the following series of non-income and income adjustments in a cumulative manner. Adjustments 2 and 3 are new to the methodology this year. As such, the participation rates are not comparable to participation rates presented in a prior paper covering 2020 (Coleman et al., 2024).

1. **Change filer from head to spouse adjustment:** The NBER TAXSIM model designates the head of a household as the primary filer, which may not always reflect actual filing behavior. This adjustment moves the modeled tax return information from the head to the spouse if the spouse filed but the head did not. If both the head and the spouse filed different returns, the modeled return information is moved from the head to the spouse when the spouse claimed CTC/ACTC, and the head did not. If both the head and spouse filed different returns, and both claimed CTC/ACTC, the modeled return information is moved from the head to the spouse when the spouse claimed the qualifying dependent and the head did not. If both filed and neither claimed CTC/ACTC, but a qualifying dependent appears on the spouse's return, the modeled return information is moved to the spouse.

[4] NBER TAXSIM is a collection of programs maintained by the National Bureau of Economic Research (NBER) that implements a model of federal and state income tax systems. Specifically, we use TAXSIM 35. More information about TAXSIM can be found at the NBER website < https://www.nber.org/research/data/taxsim>.

[5] Since the CPS ASEC collects data at the individual level, we aggregate the input variables to the tax unit level before running TAXSIM.

[6] TAXSIM 35 cannot be used to determine CTC and ACTC eligibility and amount for the 2021 tax year since it includes the Credit for Other Dependents (ODC) with the CTC/ACTC amount, and it can only take age information for up to 4 dependents in a tax unit.

2. **Adjustment for qualifying dependent (QD) claimed outside CPS ASEC households:** This adjustment removes dependents who are not claimed by any CPS ASEC household out of the analysis sample if the QD is claimed on a tax return associated with a filer not included in the CPS ASEC sample. If the original tax unit no longer has a QD after the adjustment, the tax unit is no longer considered in the set of eligible tax units.
3. **Check qualifying dependent Social Security Number:** All qualifying dependents must have a valid Social Security Number (SSN) for CTC/ACTC eligibility. This adjustment checks that each qualifying dependent has a valid SSN. The check uses the Social Security Administration's Numident file to search for a valid SSN.[7] If a qualifying dependent does not appear in the Numident file, we assume the dependent has a valid SSN if they are reported as a citizen in the survey.[8] Dependents without a valid SSN are removed from the analysis sample.
4. **Qualifying dependent (QD) adjustments for cohabiting partners, multi-generational families, and all others:** This adjustment alters the number of qualifying dependents assigned to tax units and modeled filers by comparing the number of qualifying dependents reported on the survey to the number reported on the tax return. Important to this process is the creation of a "QD unit" within a household. A "QD unit" contains all individuals in the household who could share a qualifying dependent. We make the simplifying assumption that any dependent identified by the model as a qualifying dependent could be claimed by any relative, meaning a "QD unit" consists of all related individuals in the household. We also consider non-married partners (and their children) as part of the householder's "QD unit" (i.e., they are treated as relatives of the householder).

   We adjust "QD units" based on the presence of "surplus" and "deficit" dependents. When a filer is modeled to have more qualifying dependents than they actually claimed on their return (i.e., more dependents in CPS ASEC than claimed on return), they have "surplus" children. When a filer claimed more dependents than modeled, they have "deficit" children. Non-filers are considered to not have claimed any dependents, so are classified as having a "surplus" if they are modeled to have any qualifying dependents. Adjustments are only made to units with at least one "surplus" qualifying dependent—units where, according to the CPS ASEC, there is at least one more qualifying dependent present than was actually claimed on the return. Surplus and deficit units are handled as follows based on filing, surplus, and deficit status:

   - **Filer with a surplus:** The number of qualifying dependents is reduced to the number of qualifying dependents claimed on the return. The filer's eligibility and credit amount are re-

[7] The CPS ASEC does not collect respondent or dependent Social Security Number.

[8] This requirement typically applies to younger qualifying children who may not appear in the Numident file due to lags in file processing.

estimated using TAXSIM based on the reduced number of qualifying children.

- **Non-filer with a surplus:** The number of qualifying dependents is reduced by the amount necessary to meet the needs of any deficit filers in the household (i.e., units in the household that claimed more qualifying children than modeled for them). In households comprised of both filers and non-filers with surplus dependents, deficits are first met by taking surplus dependents from filers followed by non-filers.

  As an example, if a household has a filer with 1 surplus dependent, a non-filer with 2 surplus dependents, and a filer with 1 deficit dependent, the dependent needed to make up the deficit comes from the filer with a surplus dependent (reducing the number of qualifying dependents for that filer by 1) rather than the non-filer. If the filer had a deficit of 2 dependents (rather than 1 deficit dependent), the filer with 1 surplus dependent still has that number reduced by 1, and the non-filer will have their number reduced by 1. If dependents change during this process, eligibility and credit amount are re-estimated using TAXSIM based on the new number of qualifying dependents.

- **Filer with a deficit:** The number of qualifying dependents is increased to the number of dependents claimed on the return. If the number of surplus dependents in the unit is less than the number of deficit children, deficits are met only to the extent a surplus allows.

  As an example, if a household has a filer with 2 deficit dependents and another filer with only 1 surplus dependent, the deficit filer's number of qualifying dependents is increased by 1. If dependents change during this process, eligibility and credit amount are re-estimated using TAXSIM based on the new number of qualifying dependents.

The qualifying dependent adjustments described above apply to all "QD unit" groups which include cohabiting partners, multi-generational families, and all others.

5. **Income adjustment:** The final adjustment helps correct for potential income under-reporting in the CPS ASEC and uses the best measure of wages to estimate CTC/ACTC eligibility. All CPS ASEC wages are replaced with W-2 wages. Single individuals with no W-2 wages receive a wage value of 0 unless they filed a 1040 return, in which case they receive the amount of wage and salary income from the 1040 return. For married couples, 1040 wage and salary income is used only if W-2 wages are not found for either spouse. We use any modified wages to adjust modeled earnings and modeled adjusted gross income.

   We also perform an additional income modification for investment income. If the investment income on the 1040 return exceeds the modeled investment income, the modeled income is replaced with the 1040 investment amount and modeled adjusted gross income is recomputed to

account for the higher investment amount. Similarly, if the non-wage non-investment portion of adjusted gross income is higher on the 1040 return than the corresponding portion of modeled adjusted gross income, the modeled adjusted gross income is recomputed to account for the higher 1040 amount. Following these income adjustments, eligibility and credit amount are re-estimated using TAXSIM with the new wage and income amounts.

Weighting Procedure

The CPS ASEC includes supplement weights designed to sum to the total civilian, non-institutionalized population in the United States. Individuals who are not assigned a PIK are removed from the analysis.[9] Removing these individuals from the analysis potentially causes the sample to no longer be representative, and the supplement weights will not sum to the total population. To address these concerns, we assume PIK non-assignment is missing at random conditional on race, ethnicity, age, and marital status and inflate the supplement weights for the individuals that are assigned a PIK. The factors used to inflate the supplement weight come from dividing the sample into groups defined by race, ethnicity, age, and marital status. After calculating factors for each group, we apply the factor to each supplement weight, creating a new set of adjusted weights used throughout the analysis.

Results for CTC and ACTC Participation for Tax Year 2021

Eligible taxpayers participated in the Child Tax Credit (CTC) and Additional Child Tax Credit (ACTC) program at a rate of 91 percent in tax year 2021. The dollar participation rate was 85 percent.[10] Socioeconomic factors important for CTC/ACTC participation span a broad range of categories. Households headed by females participate at rates higher than households headed by males (87 percent and 65 percent, respectively). Married filers participate more than single filers (95 percent and 82 percent, respectively). Tax units with one qualifying dependent participate at the rate of 87 percent, which is lower than the national rate, while tax units with two or three qualifying dependents participate at rates of 94 percent and 96 percent, respectively, which is higher than the national rate.[11]

[9] Refer to earlier section "Linking CPS ASEC to Administrative Tax Data" for a description of the PIK assignment process.

[10] Definitions of taxpayer and dollar participation rates as well as methods used to derive participation rates can be found in the Methodology Section at the end of the paper. All estimates represent the combined Child Tax Credit and Additional Child Tax Credit conditional on eligibility. Since the estimates derive from a new methodology developed for CTC/ACTC participation, they are subject to revision as the methodology is refined, especially the estimates related to unclaimed dollars. Refer to the "Limitations of Analysis" subsection in the Methodology Section for more information.
For comparison to prior tax years 2018, 2019, and 2020, taxpayer participation was 95 percent, 95 percent, and 94 percent, respectively, and dollar participation was 91 percent, 93 percent, and 92 percent, respectively.

[11] This paper uses the terms "taxpayers" and "tax units" interchangeably.

In tax year 2021, 3.1 million, or 9 percent, of eligible taxpayers did not participate. Tax units with adjusted gross income less than $75,000 comprise 82 percent of eligible non-participants while tax units with modeled credit amounts less than $4,500 together comprise 73 percent of eligible non- participants. Of the eligible non-participants, 53 percent filed a return.

Unclaimed CTC/ACTC dollars are estimated to be $26 billion which includes $6.9 billion in unclaimed dollars for non-filers, $5.9 billion in unclaimed dollars for filers who claimed no CTC/ACTC, and $13.2 billion in unclaimed dollars for under-claimants.[12]

Conclusion and Limitations of Analysis and Estimates

This paper provided estimates of the national taxpayer and dollar participation for the combined Child Tax Credit (CTC) and Additional Child Tax Credit (ACTC) in tax year 2021. In this tax year the credit was temporarily expanded under the American Rescue Plan Act. Overall, eligible taxpayers participated in the CTC and ACTC program at a rate of 91 percent while dollar participation was 85 percent. Certain limitations should be considered when interpreting these estimates, including error associated with modeling tax units, number of qualifying dependents, and credit eligibility and amount.

Modeling tax units and qualifying dependents subject to error: Modeling tax units and qualifying dependents could be subject to error due to differences in the reference period of demographic information collected in the CPS ASEC and tax returns. CPS ASEC demographic information used to form tax units is collected in February-April of each year while this information is reported on a tax return as of December 31 of the prior year. So, children present in the household on December 31 may not be present at the time of the interview February-April, and vice/versa. This misalignment in reference periods can lead to differences in the number and composition of tax units identified in the CPS ASEC when compared to the tax return. This difference is especially salient for the number and age of qualifying dependents in the tax unit, an important component of CTC and ACTC.

Modeling credit eligibility and amount subject to error: Modeling credit eligibility and amount could also be subject to error stemming from income differences between the survey and tax return. Differences in earnings and income reported in the survey and the tax return could affect modeled eligibility determination. We attempt to mitigate these differences through an income adjustment by using W-2 wages in place of CPS ASEC wages and replacing modeled adjusted gross income with adjusted gross income from the tax return when higher than the modeled

[12] Unclaimed dollars are subject to revision as the methodology is refined. Under-claiming can occur when the actual CTC/ACTC amount paid is less than the modeled CTC/ACTDC amount which can come from differing estimates of income and/or number of dependent children.

amount. This error should be reduced in 2021 because the earned income eligibility requirement was eliminated for the credit.

Conceptual differences in CPS ASEC and tax populations: Conceptual differences between the definitions of the CPS ASEC and tax populations may be responsible for aggregate differences between the two sources. The CPS ASEC target population covers the civilian, non-institutionalized population residing in the United States. The tax population will include the CPS ASEC population plus filers residing outside the United States, on military bases, and in institutional quarters. The tax population will also include filers with dependents residing outside the CPS ASEC household, which are generally not included on the household roster.

Survey non-response: The CPS ASEC suffers from growing non-response to individual earnings and income questions as well as the entire ASEC supplement. The Census Bureau imputes values to correct for earnings, income, and supplement non-response using a "hot deck" imputation procedure. This procedure assigns non-respondents with missing information values that come from respondents with similar characteristics. While the procedure performs reasonably well in the aggregate, it does not completely address non-response bias at the individual level which can affect modeled eligibility. While the analysis in this report retains imputed earnings and income values, it uses tax data to help mitigate non-response bias through an income adjustment by substituting W-2 wages for non-respondents (and respondents) and by substituting adjusted gross income from the tax return when higher than modeled adjusted gross income.

References

Coleman, Ciyata, Charles Hokayem, Sanghun Kim, Ethan Krohn, Krishnan Patel, and Dean Plueger. 2024. "CTC and ACTC Participation Results and IRS-Census Match Methodology, Tax Year 2020." CES Working Paper #CES-24-76, U.S. Census Bureau, < https://www.census.gov/library/working-papers/2024/adrm/CES-WP-24-76.html>

Raze, Kyle, Nicole Perales, and Liana Christin Landivar. 2026. "Non-Random Assignment of Individual Identifiers and Selection into Linked Data: Implications for Research." CES Working Paper #CES-26-06, U.S. Census Bureau, < https://www.census.gov/library/working-papers/2026/adrm/CES-WP-26-06.html>

Wagner, Deborah, and Mary Layne. 2014. "The Person Identification Validation System (PVS): Applying the Center for Administrative Records Research and Applications' (CARRA) Record Linkage Software." CARRA Working Paper #2014-01. U.S. Census Bureau.